

**II B.Tech II Semester Supplementary Examinations, Apr/May 2009
SOFTWARE ENGINEERING**

**(Common to Computer Science & Engineering, Information Technology
and Computer Science & Systems Engineering)**

Time: 3 hours

Max Marks: 80

**Answer any FIVE Questions
All Questions carry equal marks**

1. Discuss the key challenges facing the software Engineering. [16]
2. (a) Discuss the merits and demerits of spiral model. Differentiate it with other models.
(b) Elaborate the conditions under which spiral model can be applied. [5+5+6]
3. (a) Why CASE tools are needed for requirements management?
(b) Elaborate on traceability policies. [6+10]
4. (a) Define information hiding. Explain the relationship between information hiding and modular systems.
(b) What is functional independence? Discuss the ways of achieving it. [4+4+3+5]
5. (a) What is meant by User Interface? What are the three areas that user interface design focuses? Explain them.
(b) Discuss the importance of user interface design? [10+6]
6. (a) What is meant by Validation testing? Discuss the validation test criteria.
(b) Explain clearly the Alpha and Beta testing.
(c) What is meant by configuration review? Discuss its need. [6+6+4]
7. (a) What is meant by risk estimation?
(b) With an example explain how to develop a risk table [6+10]
8. (a) What is meant by FTR? Discuss about review reporting and record keeping.
(b) State and explain the guidelines for formal technical reviews. [8+8]

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1. (a) Discuss various management myths and realities.
(b) Elaborate on software crisis. [8+8]
2. (a) What is water fall model? How is it different from other engineering process models?
(b) Explain various types of evolutionary development. [5+5+6]
3. Using your own knowledge of how an ATM is used, develop a set of use-cases that could be used to derive the requirements for an ATM system. [16]
4. (a) Illustrate about component level design elements in detail.
(b) What is deployment diagram? Draw a deployment diagram for Safe Home system in UML. [6+4+6]
5. (a) Explain how human perception, and skill level will have influence in user interface design.
(b) State and explain the generic tasks that always performed in user interface design. [8+8]
6. (a) What is meant by Validation testing? Discuss the validation test criteria.
(b) Explain clearly the Alpha and Beta testing.
(c) What is meant by configuration review? Discuss its need. [6+6+4]
7. (a) How should we use metrics during the project itself?
(b) Discuss the direct and indirect measures of software process and product. [8+8]
8. (a) The Mean-Time-Between-Failure(MTBF)concept for software is open to criticism. Write few reasons for it.
(b) Can a program be correct and still not be reliable? Explain your answer.[8+8]

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1. Discuss the statement, "PSP and TSP are rigorous approaches to software engineering that provides distinct and quantifiable benefits in productivity and quality."
[16]
2. (a) What is non-functional requirements? Give examples for it.
(b) Explain the matrices for specifying non-functional requirements. [8+8]
3. (a) Define usecases. Explain their purpose.
(b) Discuss how sequence diagram add information to use case with an example.
[6+10]
4. What is architectural style? Discuss various categories of it. [6+10]
5. (a) What is meant by Layered architecture? Give an example.
(b) What is subsystem? Draw the subsystems in the weather monitoring system.
[8+8]
6. (a) What is a program graph? Discuss various notations of it.
(b) What is meant by Cyclomatic complexity? Discuss all the three ways of computing cyclomatic complexity. [6+10]
7. (a) Discuss clearly how to asses overall project risk?
(b) Discuss about risk components and drivers. [10+6]
8. (a) Is it possible asses the quality of software if the customer keeps changing? What it is supposed to do?
(b) Can a program be correct and still not exhibit good quality? Explain. [8+8]

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1. What is software process assessment ? Discuss different approaches to it . [6+10]
2. Discuss various evolutionary software process models in detail. [16]
3. (a) Define scenario. Elaborate on event scenarios.
(b) Discuss the event scenario, “When a card is entered, the customer’s PIN is requested”. [6+10]
4. (a) Define and explain abstraction and refinement. Also differentiate between them.
(b) What is refactoring? Why is it done? [4+4+3+5]
5. (a) Discuss the Mandel’s design principles that allow the user to maintain control.
(b) What is meant by usability? Discuss how to asses the usability? [8+8]
6. (a) What questions do black-box tests answer?
(b) What is meant by BVA? Discuss the guidelines to create BVA test cases.
(c) Discuss about software tools for test case design. [5+5+6]
7. (a) How should we use metrics during the project itself?
(b) Discuss the direct and indirect measures of software process and product. [8+8]
8. A Formal Technical Review (FTR) effective only if every one has prepared in advance. How do you recognize a review participant who has not prepared? What do you do if you are the review leader? [16]
